

Sudhanshu Pandey

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech Computer Science and Engineering	Birla Institute of Technology, Mesra	8.59 (Absolute)	2022-2026
Senior Secondary	St. Xaviers School Jaipur	89.8%	2021

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL, Java

Embedded & Systems: Serial Communication, UART, RTOS concepts, Newland T9 POS SDK, MQTT Protocol

Frameworks & Tools: Node.js, Express.js, FastAPI, Next.js, Electron.js, REST APIs, Docker, AWS (S3), Git, CI/CD, Prisma

Databases & Infrastructure: PostgreSQL, MongoDB, Microservices, Distributed Systems, Docker, CI/CD Pipelines, Redis

Core Concepts: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, System Design

Problem Solving: [LeetCode](#), [Codeforces](#), [Geeks for Geeks](#)

WORK EXPERIENCE

Paytm

Software Development Engineer

Noida, India

Jan 2026 – Present

- Developed Battery Health and Charging Analytics for Paytm Soundbox deployed across **13.1M+** IoT devices. Contributed in the Device TMS for Soundbox.
- Implemented network monitoring tool which measured the latency delay during transaction acknowledgment useful in detecting the root cause of transaction delay. Refactored the entire Refurb tool in Python fixing all legacy issues.
- Designed and implemented wired ECR (Electronic Cash Register) integration on Newland T9 payment terminals in India, enabling USB-serial communication between Android host applications and the embedded device for sale and status flows. Ensured robust request/response handling, checksum validation, and POS error codes, shipping on Linux-based T9 devices deployed across India.

Texas Instruments

Software Development Intern

Bengaluru, India

May 2025 – July 2025

- Developed an end-to-end Laptop Management Distribution system for Texas Instruments employees, automating **100%** of manual request workflows. Reduced manual processing time by **80%**, saving over **300** staff hours annually and cutting support costs by **25%**. Implemented caching and followed stateless architecture to increase scalability.
- Built an Election Feedback Summarizer feature which reduced the manual effort of HRs and senior management by **800-900** hrs.

WorldQuant BRAIN

Research Consultant

Remote

Dec 2024 – Mar 2025

- Conducted quantitative research and analysis to develop predictive algorithms for financial markets.
- Pioneered algorithmic trading strategy utilizing machine learning and statistical analysis, resulting in a **15%** decrease in portfolio volatility and exceeding benchmark returns by **8%** over a 12-month period.

PROJECTS

InstaChat – AI Automation | *Next.js, Node.js, Tailwind CSS, Instagram API, Prisma, PostgreSQL, Webhooks, REST APIs*

- Developed an end-to-end automation platform for Instagram Business accounts, enabling automated direct messages and comment responses. Currently used by business accounts with a combined reach of **200K+** followers.
- Integrated **Instagram Webhooks** for real-time event handling and leveraged the **OpenAI API** via **REST APIs** to generate intelligent, context-aware responses, significantly improving customer engagement.
- Sold the solution to an Instagram business manager, generating **20K** in revenue. Automated **10,000+** DMs and comment responses per month, improving response speed by **90%**.

Screenly – AI Screen Recording SaaS | *Next.js, Node.js, Express.js, TypeScript, Prisma, Electron.js, OpenAI, AWS (S3), Socket.io*

- Developed a full-stack SaaS platform for screen and webcam recording. Integrated **OpenAI Whisper** for AI-powered speech-to-text transcription and GPT-based summarization, deployed as a cross-platform Electron desktop application.
- Architected distributed backend services using **Node.js, Express.js, and Socket.io** for real-time video capture and streaming in a microservice architecture. Built with **TypeScript** and Prisma on Neon PostgreSQL with Zod validation. Collaborated using **Git**.

ACHIEVEMENTS

- Codeforces Specialist** (Max Rating: 1435), secured Rank 2542 in Codeforces DIV 2
- Solved **1000+** DSA problems across LeetCode, Codeforces, and GeeksforGeeks.
- Got G.P. Birla scholarship out of 5000 students (Top 0.1%)
- State level badminton player. Cleared C.A.T.C. 'A' and 'B' camp. NCC 'A' certificate holder